Programmable In-Network Timing Obfuscation for DNP3

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***Abstract*—An adversary watching a power-grid control network can tell which device it is looking at from how long that device takes to answer, because vendors implement the same function differently. Hiding that timing is challenging. The relay cannot be modified, and existing traffic obfuscation was built for a different feature: it reshapes flow-level sizes and inter-packet times, whereas the feature that identifies the device here is one interval inside a single transaction. It also hides the packets it adds behind encryption, which these plaintext networks do not have.**

To change the timing without touching either endpoint, we put the defense in the network. A programmable switch in front of the outstation holds the device's acknowledgment and its response, and releases each at a deadline computed from the same instant, the moment the outstation acknowledges the request. Neither release depends on how long the device took, so the interval an observer measures is a value the operator sets. The switch forwards the packets it received, so both endpoints run unmodified and no DNP3 byte changes.

We implement this on an Intel Tofino-1 and evaluate it against a physical SEL-751A protective relay. The experimental results show that the device's signature leaves the interval: it settles on the configured value with a spread more than two orders of magnitude smaller. Consequently, measuring the same exchange again gains an adversary almost nothing, and a classifier trained before deployment is left guessing. However, an adversary who retrains recovers much of what it lost from a second interval we do not target, and that residual bounds what the framework achieves.

***Index Terms*—traffic obfuscation, device fingerprinting, DNP3, programmable switches.**

I. Introduction

Device fingerprinting has been an essential step in cyber reconnaissance, allowing adversaries to reveal unique features of target networks and to design effective, stealthy attack strategies. These techniques are becoming increasingly critical in industrial control systems (ICSs) such as power grids, where adversaries often use IP-based control networks and computing devices within as entry points to inflict physical damage. Consequently, fingerprinting shifts the focus from visited websites and user biometric behavior to device models and types of control operations that are critical to ICS attacks. In the 2015 attack that disrupted Ukrainian power grids and the Stuxnet attack that disrupted Iranian nuclear power facilities, it is widely believed that adversaries stay in their systems for at least 6 months to perform cyber reconnaissance [1], [2].

To disrupt device fingerprinting, many studies present network traffic obfuscation [3], [4]. Device fingerprinting targeting general computing environments generally relies on network-level features such as packet size and/or inter-packet latency observed from communication patterns [5]. Consequently, existing traffic obfuscation often focuses on (i) padding and splitting network packets, which hide or change the distribution of network packet sizes [3], and (ii) delaying network packets and adding dummy ones, which disrupt the inter-packet timing pattern [4], [6]. Since manipulating communication networks can introduce runtime overhead, recent works have begun to offload traffic obfuscation onto programmable network switches, which change communication patterns at much higher line rates than CPUs [7]–[9].

Unfortunately, these methods cannot be directly applied to ICS environments for two main reasons. First, device fingerprinting methods on ICS devices rely on different features from the ones in general computing environments. Instead of using network layer knowledge, they attempt to use behavior on physical devices to reveal device model or types of operations. For example, work in [10] uses the latency between the TCP acknowledge packets and the actual response from the same device to estimate execution time there. Since each vendors may choose different materials or algorithms to implement the same control functionality, this time can be accurate in fingerprinting those devices. Second, existing traffic obfuscation is performed over encrypted channels, which are necessary to mix the obfuscated and normal traffic. Unfortunately, ICS networks still rely on unencrypted traffic due to a wide range of legacy devices, despite ICS network protocols may provide security features. Directly padding or adding dummy packets can easily reveal obfuscated packets, exposing the the underlying traffic pattern to adversaries.

In this paper, we present a framework that schedules when selected packets of an ICS transaction leave the network, so that a timing feature an observer measures becomes a value we choose rather than a property of the device. A programmable switch at the outstation edge holds the acknowledgment and the response of a transaction and releases each at a deadline it computes when the transaction begins. Because the switch forwards the same packets it received, the endpoints run unmodified firmware and every DNP3 byte reaches the master unchanged, which is what the plaintext setting requires. The framework is general over which packets are held and what the released interval should be. We demonstrate it on the two

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timing features that Formby et al. [10] use, the cross-layer response time of a poll and the master-visible timing of a control command, and we evaluate the first of these in depth. These are two case studies of one mechanism, not two systems.

This paper makes the following contributions:

- **Design a timing-obfuscation framework** that replaces a selected within-transaction timing feature with a configured value by scheduling the release of real protocol packets, and derive the coupling between the acknowledgment hold, the response hold, and the target interval that any such schedule must satisfy (Section IV).
- **Realize the framework on an Intel Tofino-1** using paired blocker and hold queues that sustain a hold until a deadline expires, with a read lane timed from the outstation's acknowledgment, a control lane timed from the request, and forwarding that stays open when a response arrives after its scheduled release (Section V).
- **Evaluate the realization against a physical SEL-751A relay** and report how closely the released interval reaches its target, what the policy can and cannot cover, the latency the mechanism adds, the hardware resources it uses, and what an adaptive attacker still learns from the features that remain (Section VI).

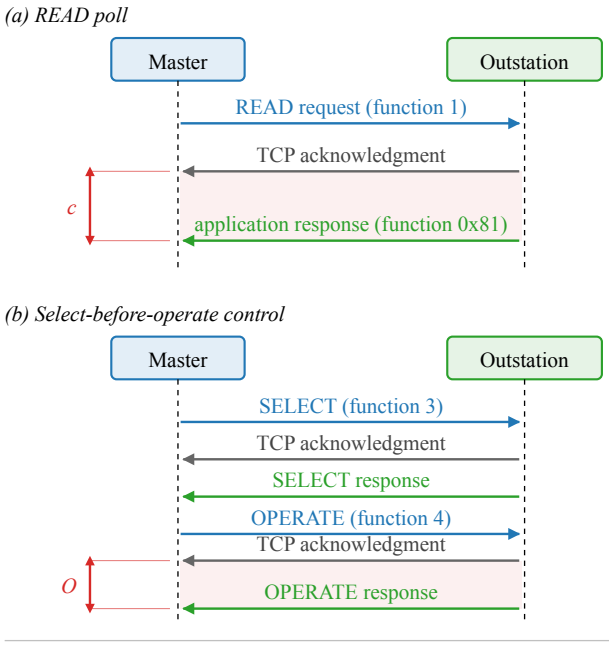
II. Background and Motivation

A defense that changes when a packet leaves has to be built on the events an observer can actually see. This section names those events for the two DNP3 transactions the paper covers. It then explains why the interval between two of them tells an observer something about the device, states which events we measured and which we did not, and describes what a programmable switch makes possible.

The DNP3 read and control transactions. DNP3 carries supervisory control and data acquisition traffic over TCP between a master and an outstation [11]. The master polls the outstation with a READ request, function code 1, and the outstation returns the requested measurements in an application-layer response, function code 0x81. A control command instead uses select-before-operate. The master sends a SELECT, function code 3, that names an output point, and the outstation returns a SELECT response. The master then sends an OPERATE, function code 4, which the outstation confirms with a solicited OPERATE response.

Because DNP3 runs over TCP, the outstation's transport layer also acknowledges each request before its application answers. Each transaction on the wire is therefore a request, a transport acknowledgment carrying no DNP3 payload, and a DNP3 response, in that order, as Figure 1 shows. We use those three words for those three packets throughout the paper.

Why the interval between them is a fingerprint. The two answers come from different parts of the outstation. Its TCP stack emits the acknowledgment as soon as the request arrives, whereas its application emits the response only after the device has done the work the request asks for. Consequently, the interval between them, the cross-layer response time or CLRT, measures the outstation's own processing rather than the



c: acknowledgment-to-response interval of the READ (CLRT).
O: master-visible OPERATE response-to-ACK interval.

Fig. 1. The two DNP3 transactions we measure, on the master-facing link. (a) A READ poll, whose cross-layer response time c runs from the acknowledgment to the response. (b) A select-before-operate control, whose master-visible interval O runs from the OPERATE's acknowledgment to its solicited response.

network path. It differs between device models and between the kinds of work a device is asked to do. For example, on our relay a control command takes 0.8 ms longer to answer than a poll, 2.937 ms against 2.116 ms, while the two requests are acknowledged 0.003 ms apart. Formby et al. used exactly this interval to tell device types and models apart in a substation network [10].

Our own measurements show the same effect on a single device. With no timing defense the median CLRT on an SEL-751A is 2.116 ms for a READ, 2.050 ms for a SELECT and 2.937 ms for an OPERATE, each with a tail reaching 83 ms. As a result, the three classes of transaction are already partly separable from timing alone (Section VI).

The DNP3 function codes travel in plaintext and already name each operation, so the timing is not what reveals that an operation happened. What it adds is a second, device-derived signature of that operation, one that needs no payload and would remain visible if the payload were encrypted.

What we observe, and what we do not. Every quantity the paper measures is one of three packet timestamps taken on the master-facing link, for the request, the acknowledgment and the response, across 132 captures. The instants at which the switch releases a packet, and every event on the relay-facing link, were not captured. Where the paper names them, they come from the configuration and we say so, and no measurement in this paper observes physical breaker motion.

What a programmable switch makes possible. A pro-

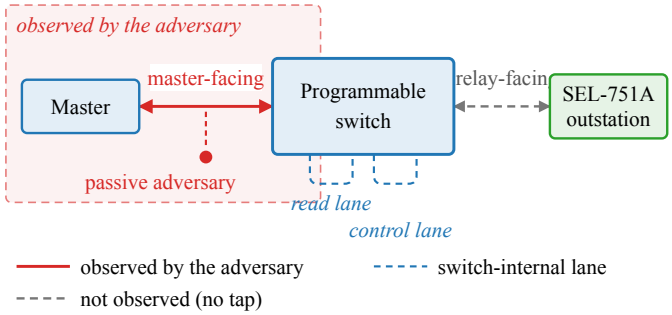


Fig. 2. Observation model. The adversary sees only the master-facing link (shaded); the relay-facing link and the switch-internal lanes (dashed) are observed by neither the adversary nor us.

grammable switch exposes a match-action pipeline, written in P4 [12], that parses headers, keeps state in registers, and places packets into output queues at line rate. The property that matters here is that such a pipeline can delay a packet and release it later without a host in the loop. Consequently, a switch placed between the master and the outstation can hold the acknowledgment and the response of a transaction and release them on a schedule an operator sets, changing neither endpoint nor any byte of the payload.

That possibility is what the rest of the paper develops. The next section states precisely which observer it is meant to defeat, and what we require of it.

III. Threat Model and Objectives

The adversary. We assume a passive observer on the master-facing link, the adversary of the fingerprinting attack we defend against [10]. Figure 2 shows where it sits. It records TCP and IP headers, segment sizes and arrival times. It does not modify or inject packets, compromise the master or the outstation, control the switch, or read the switch's internal state. It does not see the relay-facing link, which in our deployment is inside the switch with no tap on it, and it does not see physical breaker motion.

Because DNP3 carries no transport security by default, we assume the traffic is unencrypted [13], so the observer also reads all 480 DNP3 function codes per capture directly. An adversary that probes, injects, or otherwise acts on the network is out of scope. The passive observer is the setting the traffic-obfuscation work we build on also targets.

What the adversary is looking for. Reconnaissance precedes the attacks that matter here. For example, false data injection requires the attacker to know which devices it is dealing with [14]. Fingerprinting supplies that knowledge, and it does not need the payload, because the timing of the exchange carries it.

Following Formby et al. [10], two intervals are in scope. The first is the CLRT of a READ or of the SELECT phase of a control, which measures the outstation's own processing as Section II describes. On our relay the first interval's median differs by 0.9 ms between a poll and a control, against a request-to-acknowledgment interval that differs by 0.003 ms, so this is where the separation lives. The second is the master-visible interval between an OPERATE's acknowledgment and its solicited response. The adversary also sees the request-to-acknowledgment interval, whose median is 0.56 ms on this device. The mechanism does not target it, and we report it as measured.

The second interval deserves a careful statement, because it is easy to claim more for it than we can support. Formby et al. estimate the time a device physically takes to complete an operation. However, they do so from a sequence-of-events timestamp carried inside the outstation's application payload, and their alternative estimate from packet arrival times produced no usable result [10]. The interval we measure is a packet-timing feature of the solicited control response. It is not physical actuation time, and we do not measure physical actuation time anywhere. We also make no claim about the payload-timestamp feature, which a mechanism that moves packets without editing bytes could not affect in any case.

That narrower reading is still worth defending. On our relay the control lane answers 0.8 ms slower than the read lane, so the interval separates the two classes on its own, and Section VI-F shows an adversary recovering 0.651 balanced accuracy from exactly that kind of difference. RO2 therefore asks whether the interval becomes a policy value, not whether physical timing is concealed.

What our evaluation actually separates. Device fingerprinting is the application that motivates the work. What we evaluate is narrower, and we keep the two apart throughout. Our testbed has one outstation, so there is no second device to confuse it with and no device identification to measure. What we measure is whether the timing of an exchange still reveals *which class of transaction* it was, on that one device. That is a three-class problem over READ, SELECT and OPERATE, so chance is 0.333 balanced accuracy. A result here is therefore the suppression of a class-conditioned timing feature, and not a demonstration that two devices become indistinguishable.

Two adversaries, not one. A defense evaluated only against a classifier retrained on its own output answers the wrong question, so we separate two cases. The *fixed* adversary learns transaction classes from undefended traffic and applies that model unchanged once the defense is deployed. Because the profile is built before the defense is met, this is the adversary the reconnaissance setting describes.

The *adaptive* adversary collects obfuscated traffic and retrains on it. That is a strictly stronger assumption than the setting requires, and it bounds what remains learnable: it is the adversary that recovers 0.651 balanced accuracy where the fixed one is left at chance. We report both, and the second is the one that limits our claim.

Research objectives. To make the goal checkable, we state it as five properties. The evaluation answers each in turn, and we reuse the labels throughout.

- **RO1 (read timing).** The visible CLRT of READ and of the SELECT phase of SBO is a configurable policy value rather than the outstation's own processing signature,

measured as the distribution of the interval over 26,400 exchanges per arm.

- **RO2 (control timing).** The master-visible OPERATE response-to-ACK interval stays at a policy value while the relay-facing hold is drawn from a configured codebook of 2, 6 and 12 ms.
- **RO3 (timing leakage).** The timing features carry less information about the transaction class, and a fixed adversary that trained on undefended traffic is reduced to approximately three-class chance, which for READ, SELECT and OPERATE is 0.333 balanced accuracy.
- **RO4 (release policy and coverage).** The release budget, the range of policies the mechanism can realize over the 19-capture sweep, the fraction of traffic it can cover, and the residual it leaves are characterized rather than assumed.
- **RO5 (cost and protocol preservation).** We measure the latency the mechanism costs and the frames and bytes it adds, and the external DNP3 workload still completes: every request is answered and every control operation returns a success status.

What is out of scope. Three boundaries follow from the model above and hold for every result we report. The framework does not hide the DNP3 function codes, which stay in plaintext, so RO3 concerns the timing channel and not the command semantics. It forwards every packet at its original size, which is what keeps it invisible in a plaintext setting. And the relay-facing link is not instrumented, so the per-transaction hold, drawn from a configured set of 2, 6 and 12 ms, and the release toward the outstation are configured rather than observed. That is why RO2 is stated in terms of the master-visible interval alone.

RO1 and RO2 both ask for an interval the operator chooses. Whether a switch can produce such an interval at all, and what it costs to do so, is a question about timing rather than about adversaries, and the next section answers it.

IV. Design

The objectives of Section III require a selected interval to become a value the operator sets. This section explains how that is possible at all. It first states a timing model for one transaction, which is enough to see what has to change and why one arrangement of holds works where another does not. It then asks which quantities are actually free to choose and what limits each of them. It closes with the control transaction, which has to be timed from a different event, and with what the design leaves untouched. The hardware realization is deferred to Section V: a reader should be able to follow this section without knowing anything about P4.

A. A timing model for one transaction

Consider one READ transaction, drawn in Figure 3. The master sends a request. The outstation answers it twice: first its transport layer acknowledges the request, and later its application layer returns the response with the data. We write t_A for the instant the acknowledgment reaches the switch

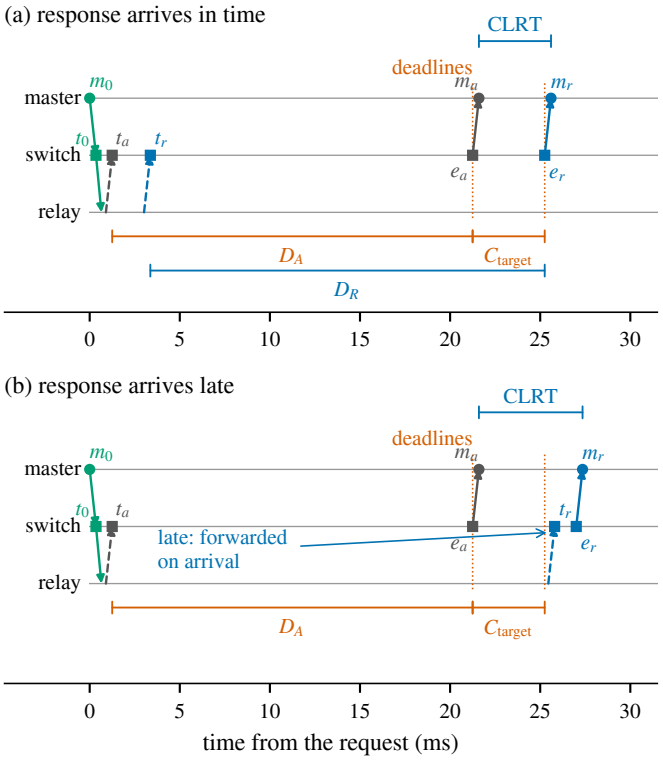


Fig. 3. Release timeline of one READ transaction. Both deadlines are computed from the acknowledgment's arrival t_a. In (a) the response arrives in time and the master observes the configured target; in (b) it arrives late and the switch forwards it at once. The response hold D_R is drawn in (a) because it is implied by the schedule, not configured. Filled circles were measured, open squares were not.

and t_R for the instant the response reaches it. The switch forwards each packet onward at some later instant, e_A and e_R respectively. The master therefore observes the interval $e_R - e_A$, up to the propagation and capture delay of the two packets on the link between them.

On our relay that interval has a median of 2.116 ms for a poll and a spread of 2.6 ms, and it is what carries information about the device:

$$\text{CLRT}_{\text{original}} = t_R - t_A, \tag{1}$$

the time the outstation itself spends between acknowledging the request and producing the answer. It is the feature Formby et al. use to tell device models apart [10], and Section II explains why; here it is simply the quantity we intend to remove.

To change what the master sees, the switch holds each packet. We call the two holds

$$D_A = e_A - t_A, \qquad D_R = e_R - t_R, \tag{2}$$

the *acknowledgment hold* and the *response hold*. Substituting the two definitions into the interval the master observes gives the identity that governs the whole design,

$$\text{CLRT}_{\text{new}} = e_R - e_A = \text{CLRT}_{\text{original}} + D_R - D_A. \tag{3}$$

Equation (3) makes the difference between two kinds of defense concrete. If the two holds are constants, their difference is a constant, and the observed interval is the original interval moved by that constant. The distribution slides along the axis and keeps its width, which on our relay means a spread of 2.6 ms wherever it is moved to. Consequently, an observer who has seen the device before can still recognize it by the shape. To remove the device's own contribution the holds cannot both be constants: D_R must absorb the variation in $\mathrm{CLRT}_{\mathrm{original}}$.

To remove the device's contribution, we compute both releases from the same instant, t_A. When the acknowledgment arrives, the switch computes both release instants immediately,

$$e_A = t_A + D_A, \qquad e_R = t_A + D_A + \mathrm{CLRT}_{\mathrm{target}}, \tag{4}$$

where $\mathrm{CLRT}_{\mathrm{target}}$ is the interval we want the master to observe. Because both release instants derive from t_A and neither derives from t_R, the term $\mathrm{CLRT}_{\mathrm{original}}$ cancels in (3) and the observed interval is the configured one. Consequently, the response hold that results is not chosen but implied,

$$D_R = D_A + \mathrm{CLRT}_{\mathrm{target}} - \mathrm{CLRT}_{\mathrm{original}}. \tag{5}$$

Equation (5) is the reason the three quantities cannot be picked independently. Once the acknowledgment hold and the target are fixed, the response hold follows from whatever the device happened to do on that transaction. Consequently, it varies from one transaction to the next exactly as much as the device does.

This holds only while the response exists in time to be released on schedule. The switch can delay a packet but cannot emit one it has not received, so (4) requires

$$t_R \le t_A + D_A + \mathrm{CLRT}_{\mathrm{target}}. \tag{6}$$

Note that the acknowledgment may leave before the response arrives. That ordering is normal and does not by itself imply a lost packet. When (6) fails, the deadline is already in the past when the response appears and the switch forwards it immediately. The master then observes an interval larger than the target, as in Figure 3(b). In our campaign 29 of 29,040 read-lane exchanges answer too late to be scheduled. We report them rather than excluding them, because they are the boundary of the mechanism and not a defect of the measurement.

B. Choosing the holds and the target

The model leaves three quantities to decide: the acknowledgment hold D_A, the target $\mathrm{CLRT}_{\mathrm{target}}$, and, through (5), the response hold D_R that follows from them. Each is limited by a different constraint.

The acknowledgment hold is limited in principle by the transport. Holding an acknowledgment delays the feedback the master's TCP uses to decide that a segment is lost. Consequently, a hold approaching the master's retransmission timeout, which RFC 6298 [15] floors at 1 s, would make the master resend a request the outstation had already answered.

Two things make this bound less tight than it first appears. The timer is adaptive. RFC 6298 [15] computes it from the round-trip times the stack has itself observed. A hold applied uniformly is therefore folded into those samples and raises the timeout rather than racing it. The standard notes the same effect from the attacker's side, as a way to inflate a sender's timer [15, §6]. What that argument does not cover is a hold that varies between transactions, which enters the variance term instead. Our policy holds by a constant, which is the case the adaptation absorbs.

In practice the transport bound is not the one that binds here. The mechanism sustains a hold with a finite resource, described in Section V, and cannot hold a packet past a horizon H that the control plane computes from it. For the configuration we evaluate, H is 30.8 ms.

Subtracting a conservative bound on the outstation's own acknowledgment latency, the measured detection and release terms, and a safety margin leaves an admissible budget of 24.8 ms. The control plane enforces that budget before it installs a policy. That ceiling is an order of magnitude below any retransmission timeout the master could plausibly use, so the mechanism reaches its own limit long before the transport notices. We report the margin we observed in Section VI-D. However, we do not claim the tested hold is universally safe: the master's realized timer was never read back, and it is the one measurement that would settle the transport question directly.

The target is limited by the operation, and here we could not find a standard that fixes it. We looked for a communications requirement that applies to a DNP3 poll or a select-before-operate control on a distribution relay. The only numeric deadline we could verify from primary documentation is the relay's own select-to-operate timeout, whose documented default is 1.0 s. Timing classes defined for other protocols govern those protocols' own messages and do not transfer to a DNP3 poll merely because the numbers are convenient.

We therefore treat the target as a deployment choice rather than a compliance question. We state the value we tested, 4 ms, and report the latency the mechanism adds, about 23 ms per exchange, so an operator with a requirement of their own can check it against those numbers.

The acknowledgment hold and the target together set what the mechanism costs. Their sum

$$D = D_A + \mathrm{CLRT}_{\mathrm{target}} \tag{7}$$

is the release budget: by (6) it decides which transactions the switch can place on the schedule at all. A larger budget covers more of the device's own response-time distribution and costs more delay on every transaction it covers: at $D = 24$ ms it covers 99.9% of them for about 23 ms of added latency.

Note that D is the budget, not the delay added to each exchange. A transaction the outstation answers quickly is held longer than one it answers slowly, so the added delay is at most D rather than equal to it.

Because the budget and the observable are different quantities, an operator can fix the cost and still choose the feature.

Holding D at 24 ms, we installed targets from 1 ms to 22 ms and the master observed each one; the delay the mechanism added stayed the same across all of them. Section VI-D reports that sweep and the point at which the realized hold stops tracking the requested one.

The framework and the policy it carries are separate. The model above says nothing about what $\mathrm{CLRT}_{\mathrm{target}}$ should be. We chose a constant target, because a constant removes the device's variation entirely and is the simplest policy to state and to check. A schedule could instead draw the target from a distribution, or make it a function of the operation, and the same mechanism would carry it. We implement and evaluate the constant target, and we do not claim evidence for the others.

C. The control transaction is timed from the request

A select-before-operate control command differs in one way that matters here: the framework holds the command itself before it reaches the relay, so that the moment the master issued the command is separated from the moment the relay acts on it. The outstation's acknowledgment cannot be the reference instant here, because it does not exist until after the command has been released. Any deadline measured from it would carry the length of the command hold into the interval the master observes.

To keep the hold out of what the master sees, we time the control lane from the request. When the OPERATE arrives at T_0, the switch releases it toward the relay after an internal delay J. It places the acknowledgment and the solicited response the master sees at configured offsets from T_0,

$$e_{\mathrm{ack}} = T_0 + A, \qquad e_{\mathrm{or}} = T_0 + R. \tag{8}$$

Consequently, the interval the master observes is

$$O = e_{\mathrm{or}} - e_{\mathrm{ack}} = R - A, \tag{9}$$

in which J does not appear. Consequently, the relay-facing release time can vary while the master-facing interval does not, and an observer cannot subtract a known offset to recover it.

The control plane checks that A exceeds the largest admissible J plus the outstation's own acknowledgment latency before it installs the values, since a deadline the switch cannot honor would fail open rather than protect. In our experiments D_A is 20 ms, $\mathrm{CLRT}_{\mathrm{target}}$ is 4 ms, and $R - A$ is also 4 ms, so the two lanes present the same value to the observer.

D. What the design does not change

The mechanism replaces one interval per lane and leaves the rest of the exchange as it was. The DNP3 function codes remain in plaintext and the direction of every packet remains visible. Consequently, an observer can still tell a poll from a control command by reading the packet. The request-to-acknowledgment interval still contains the outstation's own acknowledgment latency, displaced by the constant D_A but not reshaped by it, so whatever that interval reveals it continues to reveal.

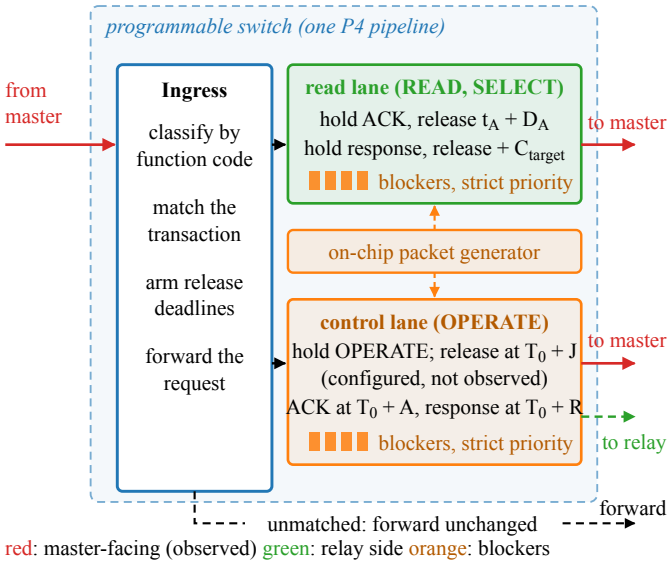


Fig. 4. How one pipeline realizes the schedule. Ingress classifies each transaction and computes its release deadlines. Blocker packets (orange) drained ahead of the held packet sustain a hold until its deadline expires. Red is master-facing and observed; the green dashed release toward the relay is not.

The switch forwards the packets it received without padding them, injecting new ones, or editing any DNP3 field, which is what the plaintext setting requires and which also means the mechanism changes no size or ordering feature. Section VI-F reports what an attacker who is allowed to retrain on the obfuscated traffic still recovers from the features that remain.

V. Implementation

Section IV says when each packet should leave. A forwarding pipeline has no way to wait: a P4 program runs to completion per packet, can read a clock, but cannot sleep or schedule a callback. This section explains how the switch releases a packet at a chosen instant anyway.

A. Holding a packet with queues

What the switch does have is a traffic manager with several queues per port and a strict-priority scheduler, the primitive with which SP-PIFO approximates programmable scheduling on today's hardware [16]. A scheduler is a device for deciding what does *not* leave. To hold a packet without a timer, we use that property.

Each lane owns two queues on one internal port, as Figure 4 shows. The *hold queue* carries the packet we delay. The *blocker queue* sits above it in strict priority and carries small packets the switch generates itself. While blockers keep arriving, the held packet is never served. The hold ends when the blocker queue runs dry. Blockers circulate on an internal loopback port and never reach a cable, so neither endpoint can observe them. Each carries a tag naming its transaction and lane, so a leftover blocker cannot delay a later transaction.

It would be natural to read this as a timer built from a count: if a blocker takes τ to drain, K blockers hold a packet for

about $K\tau$. The program does not measure time that way. To measure time without a timer, the program stores an absolute release instant when the acknowledgment arrives, quantized to a 256 ns grid, and every one of the 64 recirculating blockers compares the current timestamp against it. While the difference is negative the blocker is regenerated; once it is not, the queue empties and the held packet is served. The deadline decides when the packet leaves. The blockers only have to keep the queue non-empty until then, so the reservoir is sized for continuity rather than for duration. This is what makes the coupling of Section IV-A implementable: the response's release instant is computed at the acknowledgment, so the resulting hold absorbs whatever the outstation does without the switch ever measuring it.

B. One READ transaction

The request arrives, a table classifies it by function code, and the program forwards it at once, recording that a transaction is in flight and which acknowledgment to expect. When the acknowledgment arrives at t_A, a median of 0.56 ms after the request on our relay, the program writes both release instants from that one timestamp, which is what puts them on the same instant t_A that Section IV-A requires. It does so once from an unarmed sentinel so a duplicate acknowledgment cannot arm a second deadline. The acknowledgment enters the hold queue and the packet generator fills the blocker queue. A response that arrives before its deadline queues behind the acknowledgment, and the two leave in order at their scheduled instants. A response that arrives after its deadline has nothing left to wait for, so the program forwards it at once, as in Figure 3(b). When the response has left, the transaction state is cleared.

Read and control responses can be in flight together. To keep the two independent, we give each treatment its own loopback port, its own hold and blocker queue pair, and its own registers. Sharing one hold queue would let whichever waited longer delay the other and corrupt the policy the second was meant to present.

C. Failing open

Every path through the program ends in a forwarding decision. A packet outside the protected session, an unrecognized request, an acknowledgment with no armed transaction, and a response whose transaction has retired are all forwarded unchanged. Such an exchange loses its protection but not its delivery.

Every hold is bounded by a budget on blocker passes, which releases the packet and clears the state if the reservoir has not drained within the horizon H of Section IV-B, 30.8 ms for the configuration we evaluate. With the timing mode off the same program forwards everything immediately, which is how we collected the Timing OFF arm. The program holds and forwards packets and does nothing else to them, so every byte the master reassembles is the byte the relay produced.

D. Resources and testbed

The program is about 1,560 lines of $P4_{16}$ [12] for the Tofino Native Architecture, compiled with the Barefoot SDE 9.13. It occupies all twelve ingress match-action stages of one pipeline and six egress stages, with 112 tables. State is small: one-entry registers reached by stateful ALU actions, one access per packet per register, which is what the architecture permits. One 25 Gb/s port carries the master and one 1 Gb/s port the relay; two further ports are loopbacks carrying no cable and are the two lanes. The four hold and blocker queues of those two lanes are what the model of Section IV needs.

Every policy quantity is a runtime table entry. Consequently, an operator changes policy by writing entries rather than loading a program. The loaded build realizes the dual-deadline schedule only: its arming table admits that mode alone, so the acknowledgment-only and response-only limits of the model are properties of the design and not settings of this build. Configuring them left both packets unheld, and the measured interval was the outstation's own.

The control plane, about 890 lines of Python, brings up ports and queues, installs the packet-generator configuration, writes the timing parameters and reads them back. It also refuses configurations the data plane could not honor. It checks the admissible budget of Section IV-B and, on the control lane, that A exceeds the largest admissible J plus a bound on the outstation's acknowledgment latency. The outstation is an SEL-751A feeder protection relay, one of the DNP3 device classes whose timing Formby et al. fingerprint [10]; the master issues READ and select-before-operate operations to isolated control points through a guarded driver, so no control operation in the evaluation moves a breaker.

VI. Evaluation

In this section, we evaluate the framework against the five objectives of Section III, RO1 to RO5, on the physical testbed of Section V.

A. What we measured, and where

Evaluation Cases. We compare two arms of the same program, 31,680 exchanges each. In *Timing OFF*, the switch forwards the outstation's acknowledgment and response as they arrive; in *Obfuscated*, it holds them and releases them on the deadlines of Section IV. The two arms differ only in the release schedule. The switch is the only path between the master and the relay. Because all timestamps are taken on the master-facing link, every interval reported in this section is one a passive observer on the control network can record.

Concretely, the interval we report is the time between the arrival of the transport acknowledgment and the arrival of the application response, both observed on the master's own interface. The switch's internal instants, when the outstation's packets reach it and when it releases them, are on links that were not captured, so they are recovered from the program rather than measured.

Dataset. We report per DNP3 exchange, i.e., one request, the transport acknowledgment that follows it, and the application response.

The campaign is 22 grouped collection runs in one approximately five-hour window, with six captures of 480 exchanges per run and 132 captures in total. It contains 63,360 exchanges, 31,680 per arm, of which 26,400 are READ, 2,640 SELECT, and 2,640 OPERATE in each arm. Every request was answered, with no retransmission and no malformed frame, and every capture holds 1,448 frames and 130,708 bytes. To characterize the release policy itself, we collected a separate sweep on the same testbed. It holds 5,860 further exchanges over 19 captures: 16 configured release policies of the dual-deadline mode the loaded program realizes, and three control captures taken with the timing mechanism disabled.

B. Does the released interval reach its target? (RO1)

Effectiveness in RO1. We achieve RO1 if the read-path schedule of (4) replaces the outstation's own CLRT with the policy value. We compare the CLRT of READ and of the SELECT phase of SBO between the two arms over all 22 grouped runs as full empirical distributions. The shape of the Timing OFF distribution is what can make a single observation informative to an adversary.

Figure 6 shows the two READ distributions directly. Before obfuscation the interval is broad and multi-modal, spanning 1.01 to 83.46 ms with distinct peaks near 1.2, 2.1 and 4.0 ms. After it, 99.9% of transactions fall within 0.10 ms of the configured target. The sample variance falls from 6.808 to 0.394 ms^2, a reduction of 94.2%, and the sample standard deviation from 2.609 to 0.628 ms. The zoom panels show what the full-range view cannot. The released interval is not a single value but a narrow distribution about the target, and the Timing OFF arm places only 13.5% of its transactions anywhere in that window.

In Figure 5(d), we show the median CLRT under Timing OFF is 2.116 ms for READ and 2.050 ms for SELECT, with interquartile ranges of 2.778 ms and 2.697 ms. Under the mechanism, both medians are 4.000 ms and the interquartile range of each falls to 0.006 ms, a reduction of more than two orders of magnitude in spread.

Figure 5(a, b) shows why the median alone understates the change. The Timing OFF distributions rise in a small number of discrete steps, so a single observation can be informative about the class. The obfuscated distributions are a step at the scheduled release. This is because the release instant is computed from the outstation's acknowledgment and a fixed offset, and thus, it no longer depends on how long the device took.

The change is a replacement, not a shift. The standard deviation falls from 2.609 to 0.628 ms for READ and from 2.267 to 0.024 ms for SELECT, variance ratios of 0.058 (95% CI [0.019, 0.101]) and 0.0001 (95% CI [1.4×10^{-5}, 3.4×10^{-4}]) under a session-level bootstrap.

A defense that merely shifted the interval would keep the Timing OFF spread, because translating a sample leaves its

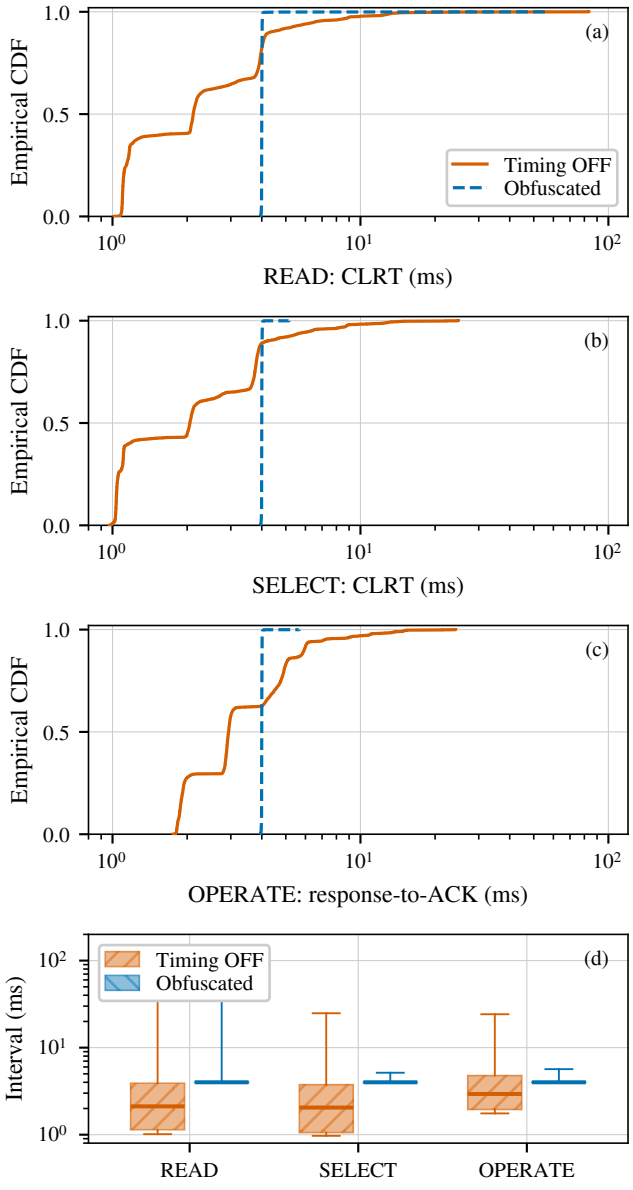


Fig. 5. Measured interval per class over 22 grouped runs. (a) READ and (b) the SELECT phase of SBO report the CLRT; (c) OPERATE reports the master-visible response-to-ACK interval, timed from the request. (d) The same measurements as quartiles. The abscissa is logarithmic and spans the full support.

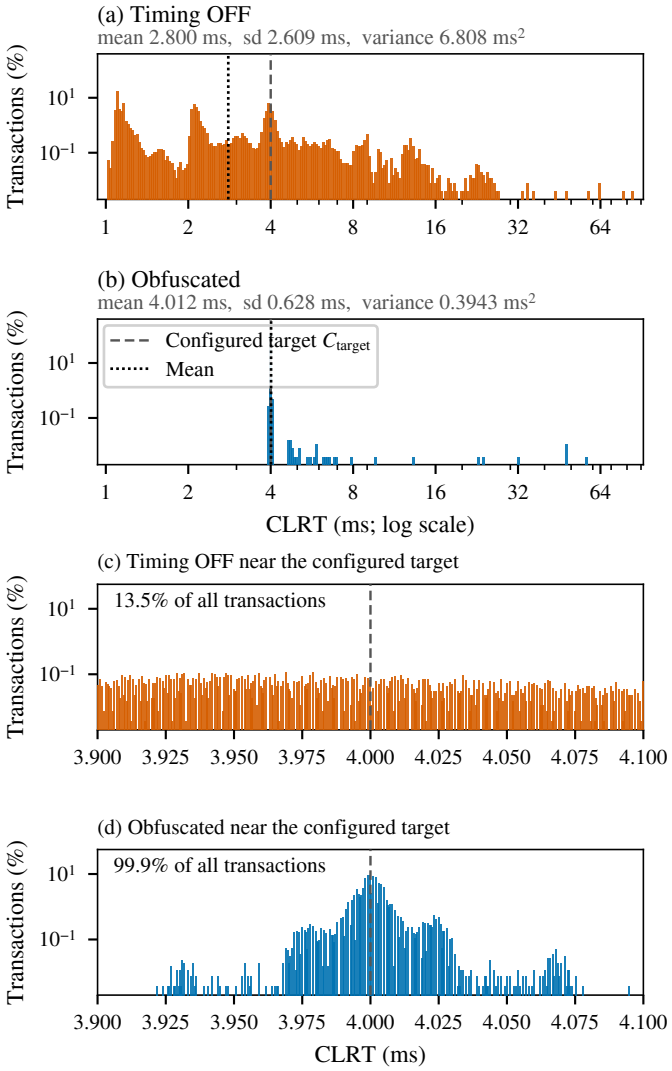


Fig. 6. Measured READ CLRT before and after obfuscation, as a percentage of each arm's own transactions. (a) and (b) give the full range on logarithmic axes and common bins; (c) and (d) zoom on 0.10 ms either side of the target. The dashed line is the configured target, the dotted line the measured mean.

variance unchanged: the counterfactual of the Timing OFF distribution translated onto the protected median retains a standard deviation of 2.609 and 2.267 ms, one to four orders of magnitude above what we measure. That counterfactual is analytical, computed from the Timing OFF samples themselves. The loaded program has no shifting mode, so it is not a measured arm and we do not present it as one. The spread collapses because both releases are computed from the acknowledgment's arrival, as Section IV sets out. A shift keeps the shape and moves it; the measurement concentrates instead.

Low spread at the wrong interval would not be normalization, so we also report where the distribution sits: both medians are the configured 4.000 ms, and the interquartile range of 0.006 ms is two orders of magnitude below the 1 ms step between neighbouring policy settings in the sweep of Section VI-D.

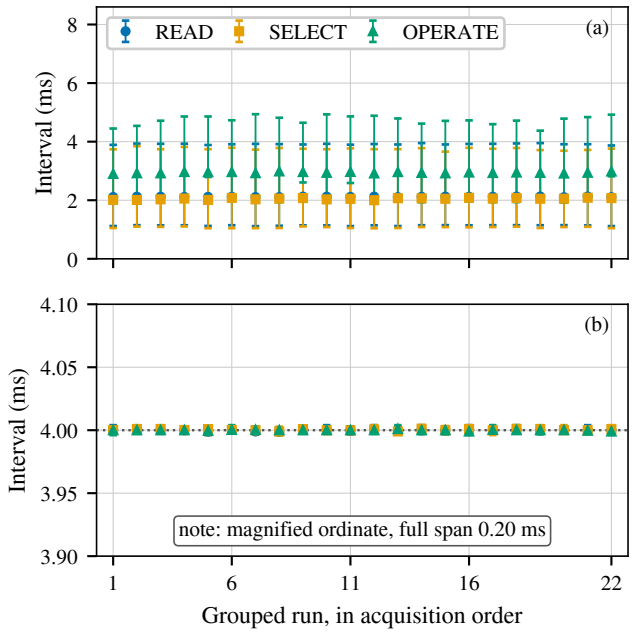


Fig. 7. Per-run medians and interquartile ranges across the 22 grouped runs, in acquisition order. *Panel (b) uses a magnified ordinate spanning 0.20 ms.* The runs are one campaign, not independent replications.

A tail remains, and it is one-sided: the largest obfuscated READ interval is 57.182 ms, and 24 of 26,400 READ exchanges depart from the scheduled release by more than 1 ms, while none falls materially below it. The asymmetry is the mechanism, not the data. A response that reaches the switch after its scheduled release cannot be moved backwards. Consequently, it is forwarded on arrival and lands above the target, and nothing can place one early. These are the coverage boundary of RO4 seen from the other side. RO1 therefore holds for the exchanges the budget can schedule: the released interval is the policy value, and the transactions that miss it are counted rather than hidden.

Stability within the Campaign. In Figure 7, we plot each grouped run in acquisition order. The obfuscated median sits at the scheduled release in every run and for every class, while the Timing OFF medians stay separated. Consequently, the effect is not an artifact of one run or of drift over the five hours.

C. Does the control lane behave the same way? (RO2)

Effectiveness in RO2. We achieve RO2 if the control-path rule of (8) holds the master-visible response-to-ACK interval at the policy value. Because the control path is timed from the request, its observable is $O = R - A$, in which the internal hold J does not appear, and the read-path budget D does not apply to it. The switch draws J per transaction from the configured codebook $\{2, 6, 12\}$ ms. As shown in Figure 5(c, d), the OPERATE median moves from 2.937 ms under Timing OFF to 4.000 ms under the mechanism, with the interquartile range falling from 2.827 ms to 0.006 ms,

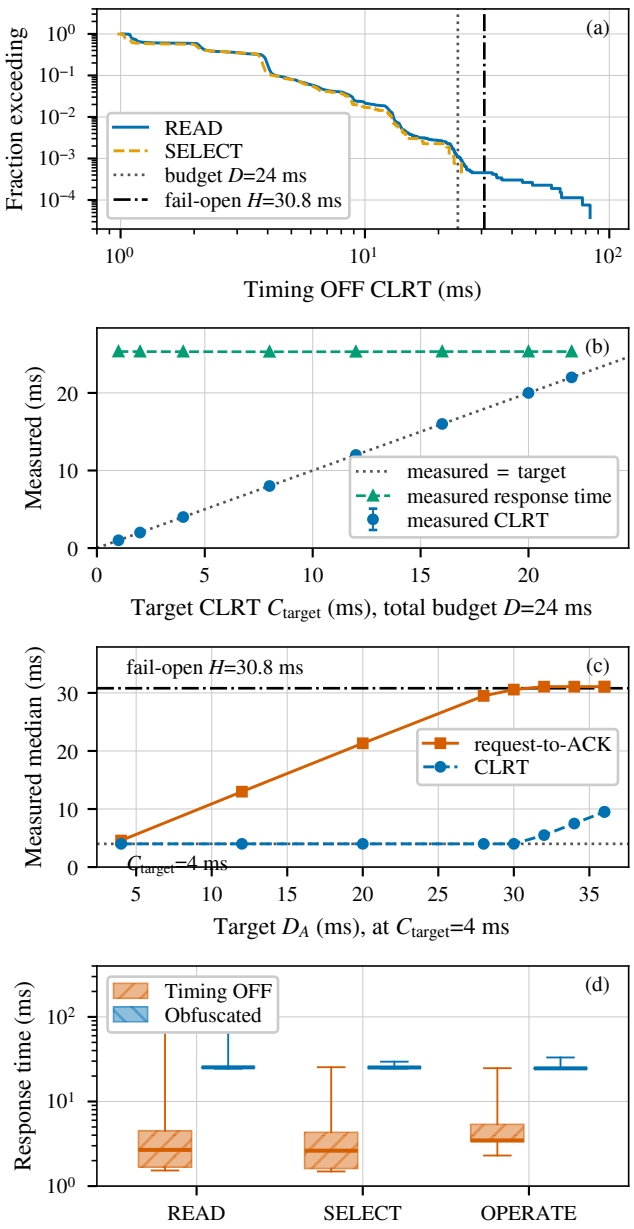


Fig. 8. The release policy is programmable and bounded on both sides. (a) Timing OFF read-lane tail against the budget D and the horizon H. (b) At $D = 24$ ms the measured CLRT follows the configured target while the response time stays fixed. (c) Raising D_A saturates near H. (d) End-to-end response time per class.

and the master-visible interval remained concentrated near the configured 4 ms policy value across all 22 grouped runs. RO2 therefore holds for the interval the master sees. Note that the relay-facing release at T_0+J was not observed (Section VI-G).

D. What range of settings does the mechanism support? (RO4)

Effectiveness in RO4. We achieve RO4 if the interval a passive observer records is a policy value the operator sets, over a usable range of settings. To measure that range on the switch itself, we installed 16 release policies in turn and ran a fixed DNP3 workload through each. Two series answer the question. The first holds the total budget at $D = D_A + \mathrm{CLRT}_{\mathrm{target}} = 24$ ms and moves the split between the two, which changes the visible CLRT while leaving the total unchanged. The second fixes the target at 4 ms and raises D_A until the mechanism leaves its operating region. Each point is the median over its READ exchanges.

In Figure 8(b), we show that the measured CLRT follows the configured target along the identity line across the whole series: targets of 1, 2, 4, 8, 12, 16, 20 and 22 ms produce measured medians of 0.998, 1.999, 3.999, 8.001, 12.001, 16.000, 20.001 and 22.001 ms, while the end-to-end response time stays between 25.30 and 25.34 ms at every point. As a result, the leaking interval and the cost of the exchange can be set independently, which is what makes the mechanism a framework rather than one fixed setting. In Figure 8(c), the request-to-acknowledgment interval tracks D_A up to 30 ms, where the measured median is 30.571 ms and the CLRT still sits at 4.001 ms.

Beyond that point, the acknowledgment interval saturates at about 31.07 ms and the CLRT can no longer be held: it rises to 5.503, 7.498 and 9.507 ms at D_A of 32, 34 and 36 ms. The reason is that the reservoir that starves the queue carries a finite pass budget. Consequently, a hold cannot outlast it, and the saturation sits close to the fail-open horizon H of Section IV. From below, the range is closed by the outstation's own response time, which the switch cannot anticipate: in Figure 8(a), the budget of 24 ms covers 99.900% of Timing OFF read-lane exchanges, and 29 of 29,040 arrive too late to be held, 28 of them READ and one SELECT. RO4 therefore holds over a range of 1 to 22 ms at a fixed budget, bounded below by the outstation's own response time and above by the 31.07 ms saturation.

E. What does the mechanism cost? (RO5)

Effectiveness in RO5. We achieve RO5 if the latency the mechanism costs and the frames and bytes it adds are measured, and the external DNP3 workload still completes. As shown in Figure 8(d), the median response time rises by 22.7 ms for READ, 22.6 ms for SELECT and 21.2 ms for OPERATE. Note that the added latency is close to, but below, the release budget, because it depends on how long the outstation itself took; it is a real operational cost that an operator would weigh against the polling period and any protocol timeout.

On the master-facing link, each capture carries 1,448 frames and 130,708 captured bytes for 480 exchanges in both arms, i.e., 3.017 frames and 272.3 bytes per exchange, identical in the two arms. In other words, the mechanism moves packets in time without adding a frame or a byte to that link. Across all 63,360 exchanges, every request was answered with a well-formed response, and all 5,280 SELECT and OPERATE exchanges returned a success status. RO5 therefore holds at a measured cost of about 23 ms per exchange and no added frame or byte on the master-facing link.

Timeout and Retransmission Headroom. Holding the acknowledgment consumes the master's timers, so we measured what it consumed. On the master-facing captures we recovered every transport-level event across all 63,360 exchanges: the request bytes were acknowledged in a separate segment every time, never piggybacked on the response, and no request was sent while earlier bytes were still unacknowledged. There was no retransmission, no reset and no duplicate acknowledgment in either arm.

The longest the master waited for its request to be acknowledged was 29.2 ms and the longest it waited for the response was 77.7 ms. We did not record the master's kernel configuration, so its actual retransmission timeout is unknown; for scale, the minimum Linux documents is 200 ms and RFC 6298 recommends a 1 s floor, and no retransmission occurred. The receive timeout in our driver is 3 s, and it bounds one read rather than the whole transaction, so it is not a transaction deadline; we report the measured completion times against it without asserting that the two coincide, which would need application-level receive tracing we did not collect.

Select validity is the other timer a held control transaction could violate, since the master issues its OPERATE only 0.19 ms after the SELECT response reaches it and the mechanism delays that response: none of the 2,640 obfuscated OPERATE exchanges was refused for a stale select. All of this characterizes this operating point on this testbed, and would have to be rechecked for a master with a shorter timeout.

F. What does the attacker still learn? (RO3)

Setup. We achieve RO3 if the timing features carry less information about the transaction class and a fixed adversary trained on undefended traffic is reduced to chance. We treat this as a three-class problem over READ, SELECT and OPERATE, for which chance balanced accuracy is one third, with the two independent intervals, request to acknowledgment and acknowledgment to response, as features; the total response time is their sum and is excluded. We estimate mutual information for the CLRT in bits and compare it against a null built by shuffling class labels within each grouped run over 1,000 permutations, the reconnaissance setting of Formby et al. [10] being the one we defend against, which preserves the per-run class counts. Every fold leaves out one grouped run, and we report the spread of the 22 held-out-run scores as a range.

Effectiveness in RO3. In Figure 9, we show what the mechanism does to the two measurable intervals before any classifier is applied. Under Timing OFF, the three classes are separated mainly along the post-acknowledgment axis, with median intervals of 2.116, 2.050 and 2.937 ms for READ, SELECT and OPERATE against request-to-acknowledgment medians of 0.555, 0.555 and 0.558 ms, which barely separate the classes at all. Under the mechanism, that axis collapses: all three medians sit at 4.000 ms, and READ and SELECT overlap. What remains is a horizontal offset, with the request-to-acknowledgment median moving to 21.336, 21.239 and 20.650 ms. This is because the read lane is timed from the outstation's acknowledgment and the control lane from the request.

In Figure 10, we show the two attacker models.

The fixed adversary trained on Timing OFF traffic reaches 0.651 balanced accuracy on held-out Timing OFF exchanges using the CLRT alone, and 0.733 using both intervals. Applied unchanged to obfuscated traffic, it falls to 0.333 and 0.334, which is chance. Mutual information between the CLRT and the class falls from 0.383 bits to 0.004 bits; the Timing OFF value lies far above its permutation null at the resolution of the test ($p = 0.001$ with 1,000 permutations), and the obfuscated value lies inside its null ($p = 0.096$). For the evaluated fixed Random-Forest attacker, then, three-class transaction identification falls to approximately chance, and the cross-layer response time stops carrying information about the class. Note that this holds for the two features we give it; the plaintext function code is outside a timing-only feature set and still names the operation.

The adaptive adversary, retrained on obfuscated traffic, remains at chance on the CLRT alone. However, it recovers to 0.651 when the acknowledgment latency is added. The reason is that the read and control paths are timed from different events, so their acknowledgment latencies differ. Therefore, an adversary willing to collect obfuscated traffic and retrain can exploit that difference; the confusion panels show its shape, with OPERATE separable while SELECT stays confused with READ. Closing this residual requires both paths to be timed from the same event, which we leave to future work. RO3 therefore holds against the fixed adversary the threat model describes, and not against one that retrains.

G. Limitations

Each limitation below bounds one of the results above, and we name which.

One Device, One Campaign. The evidence is one SEL-751A behind one Tofino-1, in one approximately five-hour campaign of 22 grouped runs. Consequently, every result from RO1 to RO5 is a statement about this device and this window. The runs are grouped collections rather than independent replications, so the spread across them is within-campaign variability.

What RO3 Establishes. The classification we evaluate separates three transaction classes on one outstation. It is not device-model identification, and with one outstation nothing here shows that two devices become indistinguishable. The result also characterizes the Random-Forest attacker we evaluated rather than every classifier, and the adaptive adversary of Section VI-F recovers 0.651 balanced accuracy from the difference between the two events, which is the bound on what RO3 achieves.

What RO2 Rests On. Only the master-facing link was captured, so the realized per-transaction hold J and the relay-facing release at $T_0 + J$ are configured rather than observed. RO2 is therefore a claim about the interval the master sees and reports no result per J.

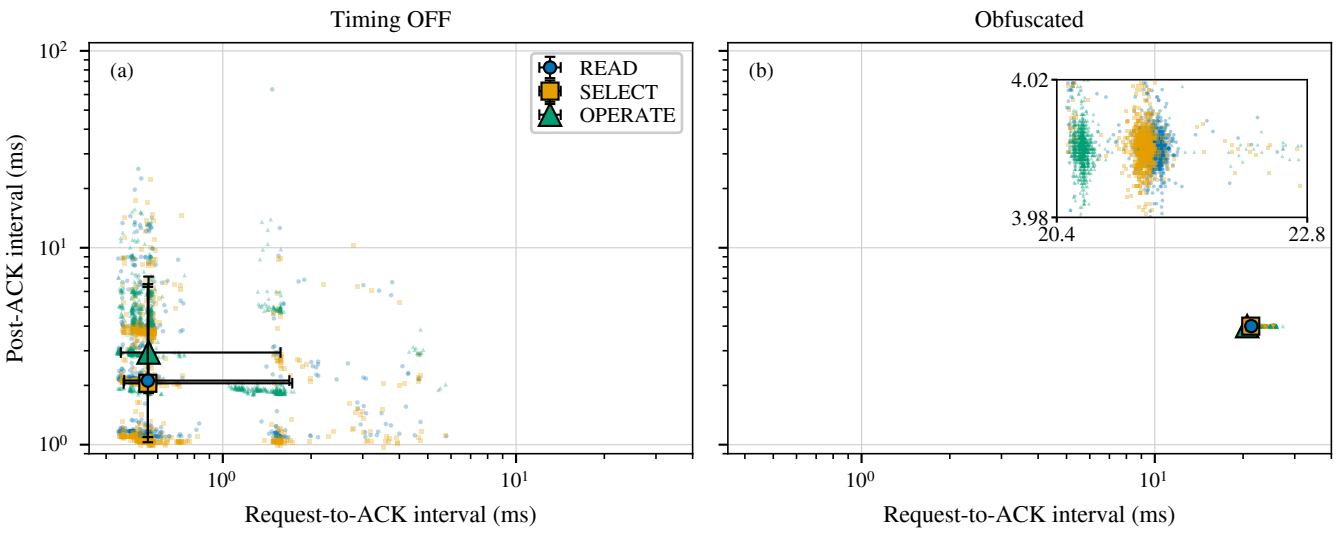


Fig. 9. The two measurable intervals against each other, (a) Timing OFF and (b) Obfuscated. Under the mechanism the vertical interval of all three classes collapses onto the policy value, while OPERATE keeps a horizontal offset because the two lanes are timed from different events. Marks are a subsample; medians and percentiles use the full dataset.

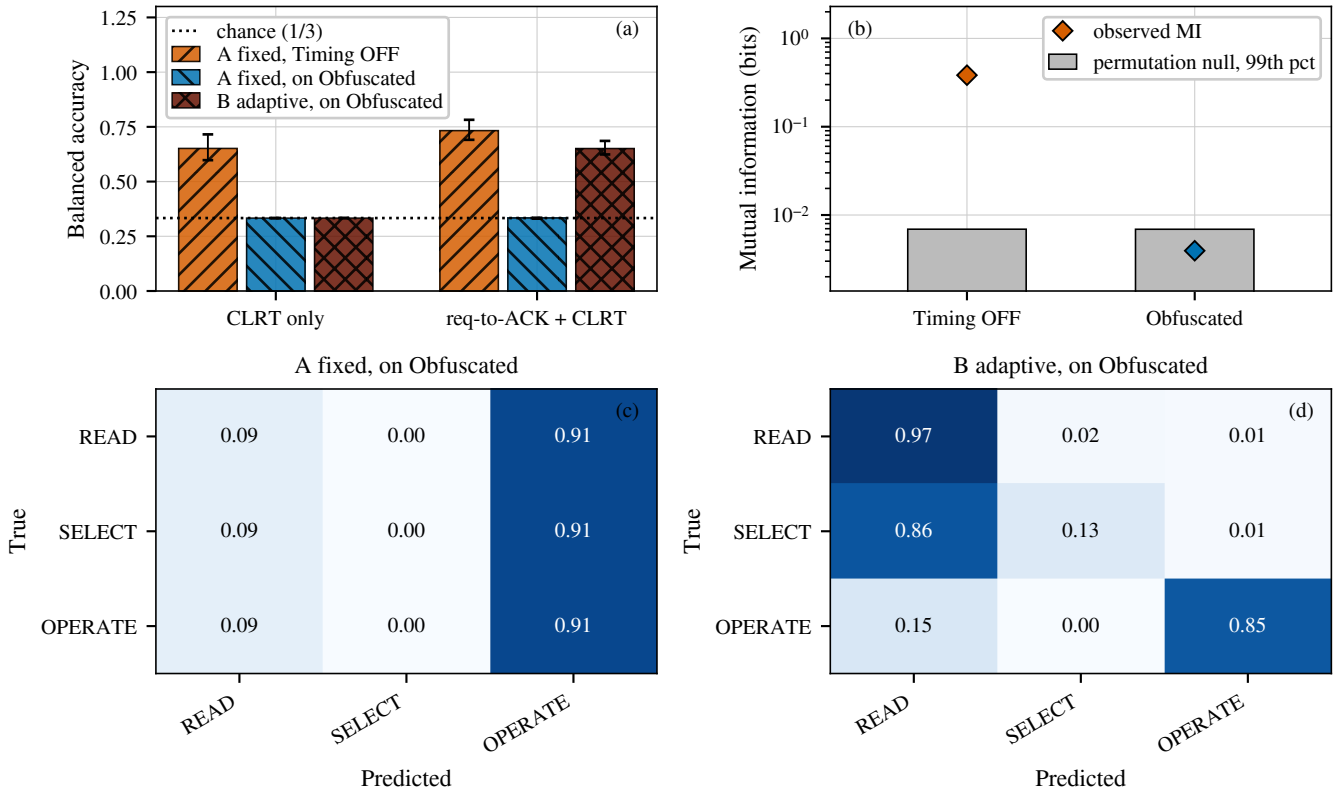


Fig. 10. Transaction-class leakage under the two attacker models, leaving out one grouped run at a time. (a) Balanced accuracy; whiskers span the 22 held-out runs and are not a confidence interval. (b) Mutual information against a within-run permutation null. (c) and (d) Row-normalized confusion using both intervals.

What RO4 Observes. The fail-open horizon H is computed in the control plane rather than enforced by the data plane, so the saturation at 31.07 ms is an observed transition and not a proof that the two coincide. The sweep offsets likewise come from the configuration table rather than from a per-point readback.

The Residual Is Not Attributed. The switch's own release instants were not captured, and the loaded program cannot supply them: the timestamp registers it declares are never written, and every write it does declare would take an ingress rather than an egress timestamp. What the measured distribution shows around the target is therefore a net difference between two releases, not a delay we can place inside the switch.

What RO5 Counts. The overhead of RO5 is measured on the master-facing link, where the switch's internal blocker traffic never appears. The zero-retransmission result is bounded the same way: because the program suppresses a response retransmission that matches an already-seen transport position, a relay-side retransmission of a held response would be absorbed inside the switch and would not reach a master-facing capture.

The Tail Is Not Eliminated. A fixed read-path release budget leaves 29 of 29,040 exchanges answering too late to be scheduled, and 24 of 26,400 protected READ exchanges more than 1 ms from the target. RO1 is a claim about the distribution, not about every transaction in it.

VII. Related Work

Device fingerprinting. Devices can be identified from what they emit, even without any access to the payload. Kohno et al. fingerprint a remote host from the skew of its clock as seen in TCP timestamps [17], and GTID fingerprints a device and its type from the timing signature of its packets [18]. These works define the threat we address; our objective, on the other hand, is to remove the timing features they rely on from the master-facing observation. Their target is the device type, whereas the classifier we evaluate separates three transaction classes on one outstation, so we suppress a feature their cross-layer response-time method uses without demonstrating that two devices become indistinguishable.

ICS and power-system fingerprinting. Formby et al. brought fingerprinting to control systems with the cross-layer response time and the physical operation time, measured at the device, of DNP3 outstations [10], and Gu et al. added device physics as a feature [19]. Jeon et al. fingerprint SCADA devices passively without deep packet inspection [20]. The reconnaissance these techniques serve enables false-data injection and process-control attacks [14], [21].

Defenses in this setting have so far worked at the content and connectivity layer: DefRec virtualizes physical functions to present deceptive content [22], RAINCOAT randomizes data acquisition and injects decoy measurements [23], and a companion testbed evaluates such anti-reconnaissance approaches on grid infrastructure [24]. Compared to these, our framework works below the semantics they reshape and complements them: it changes when the wire carries the outstation's answer, not what the answer says.

General traffic obfuscation. Traffic morphing transforms one class's packet-size distribution into another's [3], while Dyer et al. show that per-packet padding and fragmentation still leave exploitable features [4]. Tamaraw fixes packet size and rate at a bandwidth cost [6], and WTF-PAD pads inter-packet gaps adaptively [25]. These defenses are effective against the size features they target, and the attacks that follow them are correspondingly strong [5], [26].

The same size-and-timing channel identifies smart-home devices under encryption, where shaping is the countermeasure [27], and Pacer and NetShaper shape a tenant's traffic to a schedule that bounds side-channel leakage in the cloud [28], [29]. These defenses assume an encrypted payload and a cooperating end host or hypervisor that can add cover traffic. Unlike those settings, ours carries a plaintext protocol whose leaking feature is one interval inside a request-response exchange. Because the outstation cannot be changed, it adds no byte and runs entirely in the network.

Programmable data-plane traffic transformation. P4 defines the programmable parser and match-action model we use [12]. PIFO defines programmable scheduling at line rate [30], and SP-PIFO approximates it with strict-priority queues on today's switches [16], which is the primitive our reservoirs use. Ditto pads and injects chaff to a fixed pattern at line rate [7], Minos morphs and schedules traffic on a programmable switch at low cost [8], and Securitas combines fragmentation and insertion, and realizes them across several data planes [9]. NetHide obfuscates topology in the data plane against link-flooding reconnaissance [31]. Closest to our setting, Hu et al. use programmable switches to enhance industrial-protocol security [32]. Our framework differs from these in the observable it addresses, i.e., the master-visible timing of a DNP3 transaction, and in holding packets rather than adding or reshaping them.

DNP3 timing and security. East et al. catalogue attacks on DNP3 and the absence of encryption that makes its semantics visible [13], and specification-based and state-based intrusion detection has been adapted to DNP3 [33], [34]. These do not change the timing an observer sees.

On the control path we report a different quantity from Formby et al.: the master-visible OPERATE response-to-acknowledgment interval, which is a protocol-response timing feature rather than the payload-timestamp estimate of physical operation time they use [10]. Section III states why the two are not interchangeable and what we therefore do not claim.

VIII. Conclusion

How long an outstation takes to answer tells an observer which device it is, and the relay leaking it cannot be modified. This paper presents a timing-obfuscation framework that moves the problem into the network: a switch in front of the outstation holds the packets carrying the device's timing and releases each at a deadline computed when the transaction begins. Both deadlines are computed from the same instant,

the moment the outstation acknowledges the request. Consequently, the interval an observer sees is the operator's value rather than the device's.

Evaluations on a physical relay over 22 collection runs show that the device's signature leaves that interval. Its spread falls from 2.61 to 0.63 ms, and a classifier trained before deployment drops from 0.651 balanced accuracy to three-class chance. The framework protects the exchanges its budget can schedule and forwards the rest untouched.

Two bounds limit the result. An adversary who retrains recovers to 0.651 from a second interval we do not target, and the evidence is one relay, one campaign and one classifier. We also never captured the switch's own release instants, so the residual around the target is a difference between two releases rather than a delay we can place inside the switch. In future work, we plan to evaluate the framework on outstations from several vendors and to measure the release instants directly with synchronized capture on both switch links.

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